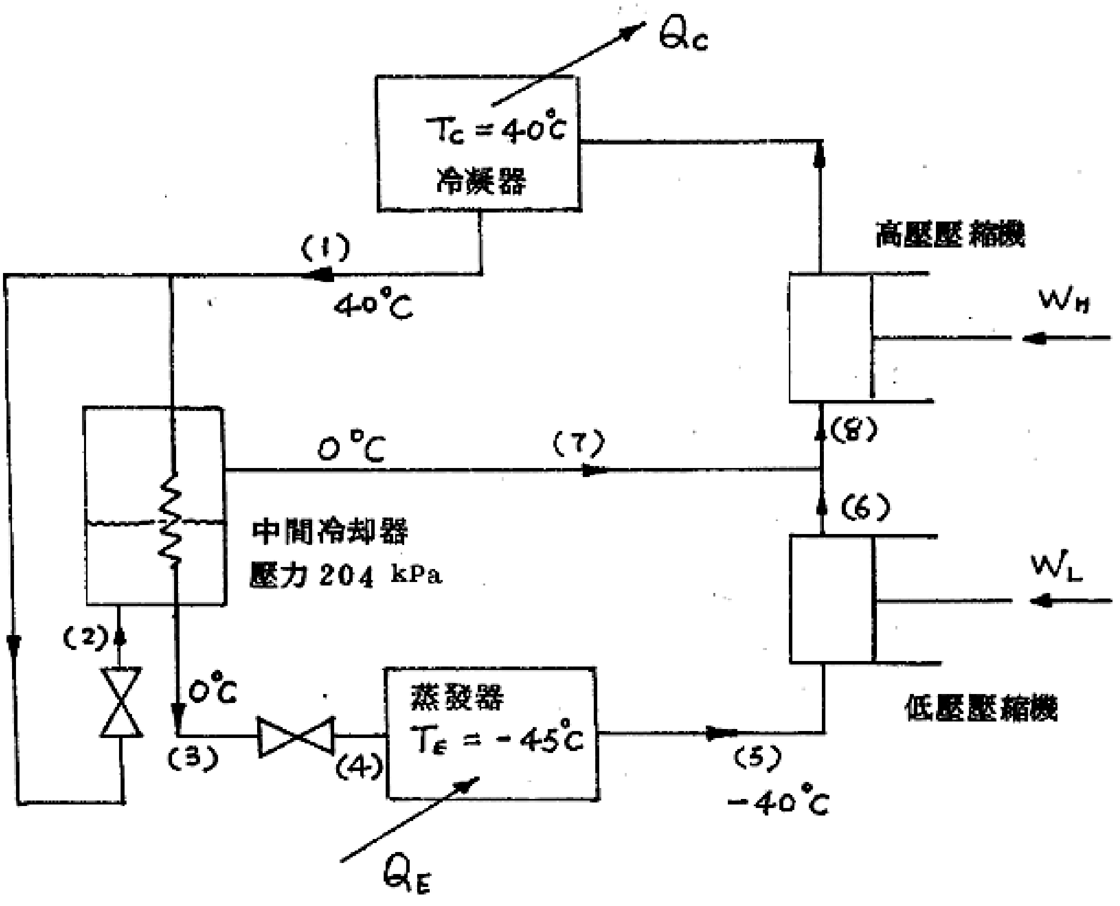
類　　科：冷凍空調工程技師
科　　目：冷凍空調原理
考試時間：二小時

一、有一固定風量雙風管空調系統（constant air volume dual duct system）其空調區（zone）之設計熱負載（heating load）爲10 kW，設計冷負載（cooling load）爲8 kW。空調區之溫度維持在25℃，冷風之送風溫度爲12℃，熱風之送風溫度爲40℃，混合空氣（回風口）之溫度爲25℃。當卸載冷却負載爲3 kW時，請計算：
㈠總風量，kg/s
㈡進入空調區之送風溫度，℃
㈢加熱盤管及冷却盤管之負載，kW　（30分）

二、茲以雙風管空調系統（dual duct system）爲例，請說明及比較固定風量系統（constant air volume）與可變風量（VAV, variable air volume）空調系統之異同。（20分）

三、用以調節冷凍能力之熱氣旁通方式其中一種爲旁通至膨脹閥與蒸發器之間，另一種爲直接旁通至壓縮機吸氣管，試：
㈠繪出該兩種系統之p－h圖。
㈡計算冷媒R－12系統當其採用第二種旁通方式（旁通至低壓管）並在下列條件下之卸載能力及旁通冷媒流量：
冷凝溫度40℃，蒸發溫度0℃，壓縮機容積效率0.8，全載時能量爲5 kW，高壓排氣之容許極限溫度爲100℃。（20分）

四、有一冷媒R－12兩段壓縮冷凍系統如圖所示：（附R－12性質表）
㈠繪出該系統之P－h圖。
㈡計算該系統之性能係數COP。（30分）



附圖及表

八十二年第一次專門職業及技術人員 律 師、會計師 / 建築師、技 師 檢覈筆試試題 3－9 全一
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類 科：冷凍空調工程技師

科 目：冷凍空調原理

冷媒R－12 P－h圖

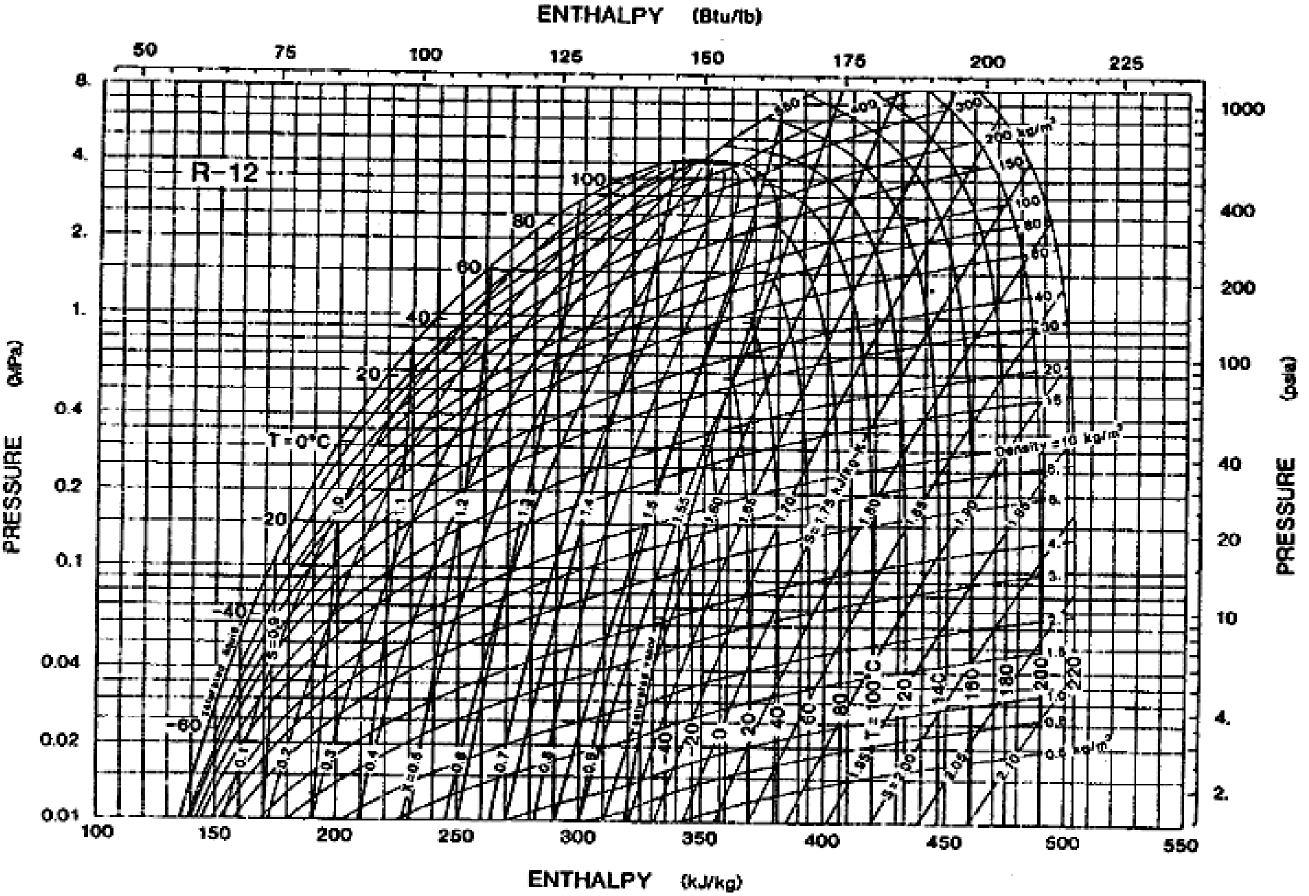

Table A-5 Refrigerant 12: properties of liquid and saturated vapor[5]

t, °C	P, kPa	Enthalpy, kJ/kg h_f	Enthalpy, kJ/kg h_g	Entropy, kJ/kg · K s_f	Entropy, kJ/kg · K s_g	Specific volume, L/kg v_f	Specific volume, L/kg v_g
-60	22.62	146.463	324.236	0.77977	1.61373	0.63689	637.911
-55	29.98	150.808	326.567	0.79990	1.60552	0.64226	491.000
-50	39.15	155.169	328.897	0.81964	1.59810	0.64782	383.105
-45	50.44	159.549	331.223	0.83901	1.59142	0.65355	302.683
-40	64.17	163.948	333.541	0.85805	1.58539	0.65949	241.910
-35	80.71	168.369	335.849	0.86776	1.57996	0.66563	195.398
-30	100.41	172.810	338.143	0.89516	1.57507	0.67200	159.375
-28	109.27	174.593	339.057	0.90244	1.57326	0.67461	147.275
-26	118.72	176.380	339.968	0.90967	1.57152	0.67726	136.284
-24	128.80	178.171	340.876	0.91686	1.56985	0.67996	126.282
-22	139.53	179.965	341.780	0.92400	1.56825	0.68269	117.167
-20	150.93	181.764	342.682	0.93110	1.56672	0.68547	108.847
-18	163.04	183.567	343.580	0.93816	1.56526	0.68829	101.242
-16	175.89	185.374	344.474	0.94518	1.56385	0.69115	94.2788
-14	189.50	187.185	345.365	0.95216	1.56250	0.69407	87.8951
-12	203.90	189.001	346.252	0.95910	1.56121	0.69703	82.0344
-10	219.12	190.822	347.134	0.96601	1.55997	0.70004	76.6464
-9	227.04	191.734	347.574	0.96945	1.55938	0.70157	74.1155
-8	235.19	192.647	348.012	0.97287	1.55897	0.70310	71.6864
-7	243.55	193.562	348.450	0.97629	1.55822	0.70465	69.3543
-6	252.14	194.477	348.886	0.97971	1.55765	0.70622	67.1146
-5	260.96	195.395	349.321	0.98311	1.55710	0.70780	64.9629
-4	270.01	196.313	349.755	0.98650	1.55657	0.70939	62.8952
-3	279.30	197.233	350.187	0.98989	1.55604	0.71099	60.9075
-2	288.82	198.154	350.619	0.99327	1.55552	0.71261	58.9963
-1	298.59	199.076	351.049	0.99664	1.55502	0.71425	57.1579
0	308.61	200.000	351.477	1.00000	1.55452	0.71590	55.3892
1	318.88	200.925	351.905	1.00335	1.55404	0.71756	53.6869
2	329.40	201.852	352.331	1.00670	1.55356	0.71924	52.0481
3	340.19	202.780	352.755	1.01004	1.55310	0.72094	50.4700
4	351.24	203.710	353.179	1.01337	1.55264	0.72265	48.9499
5	263.55	204.642	353.600	1.01670	1.55220	0.72438	47.4853
6	374.14	205.575	354.020	1.02001	1.55176	0.72612	46.0737
7	386.01	206.509	354.439	1.02333	1.55133	0.72788	44.7129
8	398.15	207.445	354.856	1.02663	1.55091	0.72966	43.4006
9	410.56	208.383	355.272	1.02993	1.55050	0.73146	42.1349
10	423.30	209.323	355.686	1.03322	1.55010	0.73326	40.9137
11	436.31	210.264	356.098	1.03650	1.54970	0.73510	39.7352
12	449.62	211.207	356.509	1.03978	1.54931	0.73695	38.5975
13	463.23	212.152	356.918	1.04305	1.54893	0.73882	37.4991
14	477.14	213.099	357.325	1.04632	1.54856	0.74071	36.4382
15	491.37	214.048	357.730	1.04958	1.54819	0.74262	35.4133
16	505.91	214.998	358.124	1.05284	1.54783	0.74455	34.4230
17	520.76	215.951	358.535	1.05609	1.54748	0.74649	33.4658
18	535.94	216.906	358.935	1.05933	1.54713	0.74846	32.5405
19	551.45	217.863	359.333	1.06258	1.54679	0.75045	31.6457
20	567.29	218.821	359.729	1.06581	1.54645	0.75246	30.7802

Table A-5 (continued)

t, °C	P, kPa	Enthalpy, kJ/kg h_f	Enthalpy, kJ/kg h_g	Entropy, kJ/kg · K s_f	Entropy, kJ/kg · K s_g	Specific volume, L/kg v_f	Specific volume, L/kg v_g
21	583.47	219.783	360.122	1.06904	1.54612	0.75449	29.9429
22	599.98	220.746	360.514	1.07227	1.54579	0.75655	29.1327
23	616.84	221.712	360.904	1.07549	1.54547	0.75863	28.3485
24	634.05	222.680	361.291	1.07871	1.54515	0.76073	27.5894
25	651.62	223.650	361.676	1.08193	1.54484	0.76286	26.8542
26	669.54	224.623	362.059	1.08514	1.54453	0.76501	26.1422
27	687.82	225.298	362.439	1.08835	1.54423	0.76718	25.4524
28	706.47	226.576	362.817	1.09155	1.54393	0.76938	24.7840
29	725.50	227.557	363.193	1.09475	1.54363	0.77161	24.1362
30	744.90	228.540	363.566	1.09795	1.54334	0.77386	23.5082
31	764.68	229.526	363.937	1.10115	1.54305	0.77614	22.8993
32	784.85	230.515	364.305	1.10434	1.54276	0.77845	22.3088
33	805.41	231.506	364.670	1.10753	1.54247	0.78079	21.7359
34	826.36	232.501	365.033	1.11072	1.54219	0.78316	21.1802
35	847.72	233.498	365.392	1.11391	1.54191	0.78556	20.6408
36	869.48	234.499	365.749	1.11710	1.54163	0.78799	20.1173
37	891.64	235.503	366.103	1.12028	1.54135	0.79045	19.6091
38	914.23	236.510	366.454	1.12347	1.54107	0.79294	19.1156
39	937.73	237.521	366.802	1.12665	1.54079	0.79546	18.6362
40	960.65	238.535	367.146	1.12984	1.54051	0.79802	18.1706
41	984.51	239.552	367.487	1.13302	1.54024	0.80062	17.7182
42	1008.8	240.574	367.825	1.13620	1.53996	0.80325	17.2785
43	1033.5	241.598	368.160	1.13938	1.53968	0.80592	16.8511
44	1058.7	242.627	368.491	1.14257	1.53941	0.80863	16.4356
45	1084.3	243.659	368.818	1.14575	1.53913	0.81137	16.0316
46	1110.4	244.696	369.141	1.14894	1.53885	0.81416	15.6386
47	1136.9	245.736	369.461	1.15213	1.53856	0.81698	15.2563
48	1163.9	246.781	369.777	1.15532	1.53828	0.81985	14.8844
49	1191.4	247.830	370.088	1.15851	1.53799	0.82277	14.5224
50	1219.3	248.884	370.396	1.16170	1.53770	0.82573	14.1701
52	1276.6	251.004	370.997	1.16810	1.53712	0.83179	13.4931
54	1335.9	253.144	371.581	1.17451	1.53651	0.83804	12.8509
56	1397.2	255.304	372.145	1.18093	1.53589	0.84451	12.2412
58	1460.5	257.486	372.688	1.18738	1.53524	0.85121	11.6620
60	1525.9	259.690	373.210	1.19384	1.53457	0.85814	11.1113
62	1593.5	261.919	373.707	1.20034	1.53387	0.86534	10.5872
64	1663.2	264.172	374.180	1.20686	1.53313	0.87282	10.0881
66	1735.1	256.452	374.625	1.21342	1.53235	0.88059	9.61234
68	1809.3	268.762	375.042	1.22001	1.53153	0.88870	9.15844
70	1885.8	271.102	375.427	1.22665	1.53066	0.89716	8.72502
75	2087.5	277.100	376.234	1.24347	1.52821	0.92009	7.72258
80	2304.6	283.341	376.777	1.26069	1.52526	0.94612	6.82143
85	2538.0	289.879	376.985	1.27845	1.52164	0.97621	6.00494
90	2788.5	296.788	376.748	1.29691	1.51708	1.01190	5.25759
95	3056.9	304.181	375.887	1.31637	1.51113	1.05581	4.56341
100	3344.1	312.261	374.070	1.33732	1.50296	1.11311	3.90280